

Erdős Problem #420

EP-420	L1 - Tractable	Ouvert	Number Theory
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Énoncé original (anglais, reproduction mot pour mot)

If $\tau(n)$ counts the number of divisors of n then let $F(f,n) = \frac{\tau((n + \lfloor f(n) \rfloor)!)}{\tau(n!)}$. Is it true that $\lim_{n \rightarrow \infty} F((\log n)^C, n) = \infty$ for large C ?
Is it true that $F(\log n, n)$ is everywhere dense in $(1, \infty)$?
More generally, if $f(n) \leq \log n$ is a monotonic function such that $f(n) \rightarrow \infty$ as $n \rightarrow \infty$, then is $F(f,n)$ everywhere dense?

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